MAH Governance Cost Mechanism: Model Mapping, Bellman Equations, and Empirical Implementation

1 Purpose

This note integrates two tasks into one coherent framework.

First, it gives a direct model mapping for the central mechanism in the MAH paper: the reform lowers the governance and compliance friction of *external commercialization* for research-originated original-drug ideas.

Second, it updates the variable definitions and empirical implementation plan so that the empirical section, the static model block, and the dynamic Bellman equations all use the same object.

The key update is the following:

$$\tau_{ext}$$

is the central primitive through which MAH enters the model. It is *not* a generic policy dummy, and it is *not* total governance cost for the whole firm. It is the *route-specific governance/compliance cost* of commercializing a research-originated original-drug idea through an external qualified manufacturer while the innovator retains authorization status.

2 What τ_{ext} is and what it is not

2.1 Definition

Define

$$\tau_{ext} \geq 0$$

as the extra organizational, contractual, and compliance cost that a research-oriented firm must bear when it keeps the project/authorization and completes commercialization through an external qualified manufacturer rather than through full internal integration.

This cost includes:

- drafting and enforcing entrustment agreements and quality agreements;
- auditing, monitoring, and coordinating the contract manufacturer;

- complying with holder-side quality responsibility and release obligations;
- managing cross-firm execution and accountability in the commercialization stage.

2.2 What it does not include

It does *not* include:

- physical manufacturing cost itself;
- R&D cost of generating ideas;
- economy-wide demand shifts;
- a generic fixed operating cost of all firms.

Thus the correct interpretation is:

$$\tau_{ext} = \text{governance/compliance cost of the external commercialization route}$$

rather than

$$\tau_{ext} = \text{all governance cost in the economy.}$$

3 Why this is the right MAH mechanism

The economic role of MAH is not that it makes scientists more productive, and not that it directly raises drug demand. Its distinctive role is organizational.

Before MAH, a research-originating firm without internal manufacturing capacity could typically commercialize an original-drug idea only by integrating downstream production or by relinquishing the project to an integrated producer. After MAH, the innovator may retain authorization while contracting with a qualified external manufacturer, under a legally recognized structure with explicit responsibility allocation and standardized compliance requirements.

Therefore, the reform changes the cost of completing commercialization *across firm boundaries*. In model language, MAH lowers the route-specific governance/compliance friction of external commercialization:

$$\tau_{ext}^{post} < \tau_{ext}^{pre}.$$

4 Where τ_{ext} enters the model

4.1 Core principle

The key modeling principle is:

$\boxed{\tau_{ext} \text{ enters only at the stage where a research-originated idea is commercialized externally.}}$

It should not directly enter:

- demand,
- scientific productivity,
- the physical production technology of manufacturers,
- generic fixed cost for all firms.

Instead, it enters the *commercialization assignment problem* for original-drug ideas.

5 Static block

5.1 External commercialization payoff for a type-B idea

Let z denote firm capability and p_m the price of qualified contract manufacturing.

For a research-specialized firm B, define the net payoff from commercializing one original-drug idea through external qualified production as

$$H_B^{out}(z, p_m; \tau_{ext}) = \lambda_B(z, p_m)\, R_B(z) - p_m - \tau_{ext},$$

where:

- $R_B(z)$ is the private value of a successfully commercialized original-drug idea;
- $\lambda_B(z, p_m)$ is the success rate of external commercialization;
- p_m is the contract-manufacturing price;
- τ_{ext} is the route-specific governance/compliance friction.

This separates two conceptually different costs:

$$\begin{aligned} p_m &: \text{market/production payment to the external manufacturer,} \\ \tau_{ext} &: \text{organizational and legal cost of executing the route.} \end{aligned}$$

5.2 Commercialization value per idea

If the external route is the only route being emphasized, define

$$\Psi_B(z; p_m, \tau_{ext}) = \max\left\{H_B^{out}(z, p_m; \tau_{ext}),\, 0\right\}.$$

If an internal integration or project-transfer route is retained, then more generally:

$$\Psi_B(z; p_m, \tau_{ext}) = \max\left\{H_B^{out}(z, p_m; \tau_{ext}), H_B^{int}(z), 0\right\}.$$

The crucial point is that τ_{ext} enters only the external route:

$$\frac{\partial H_B^{out}}{\partial \tau_{ext}} = -1, \qquad \frac{\partial H_B^{int}}{\partial \tau_{ext}} = 0.$$

5.3 Idea-generation problem of a type-B firm

Let $x_B \geq 0$ denote idea-generation intensity and let $\eta > 1$ govern convex idea-generation cost. Then the optimized current-period payoff of a type-B firm is

$$\Pi_B(z; p_m, \tau_{ext}) = -f_B + \max_{x_B \geq 0} \left\{ x_B \Psi_B(z; p_m, \tau_{ext}) - \frac{\chi_B}{\eta} x_B^{\eta} \right\}.$$

The first-order condition yields

$$x_B^*(z; p_m, \tau_{ext}) = \left(\frac{\Psi_B(z; p_m, \tau_{ext})}{\chi_B} \right)^{\frac{1}{\eta-1}} \quad \text{if } \Psi_B > 0.$$

Hence the mechanism runs through:

$$\tau_{ext} \downarrow \Rightarrow H_B^{out} \uparrow \Rightarrow \Psi_B \uparrow \Rightarrow x_B^* \uparrow .$$

6 Dynamic Bellman system

6.1 Type-B Bellman equation

Let $V_B(z)$ denote the value of operating as a type-B firm at state z. Suppose the firm can remain B, switch to A, switch to C, or exit.

Define:

$$W_{BB}(z) = \Pi_B(z; p_m, \tau_{ext}) + \beta E\left[V_B(z') \mid z, B\right],$$

$$W_{BA}(z) = -\kappa_{BA} + \Pi_A(z; w) + \beta E\left[V_A(z') \mid z, A\right],$$

$$W_{BC}(z) = -\kappa_{BC} + \Pi_C(z; p_m, w) + \beta E\left[V_C(z') \mid z, C\right].$$

Then:

$$V_B(z) = \max\left\{W_{BB}(z),\, W_{BA}(z),\, W_{BC}(z),\, 0\right\}.$$

The key point is:

$$\tau_{ext} \text{ enters } V_B(z) \text{ through } W_{BB}(z) \text{ only.}$$

That is, τ_{ext} affects the value of remaining a research-specialized firm and using the external commercialization route. It does not mechanically alter the value of becoming integrated or becoming manufacturing-specialized.

6.2 Type-A and Type-C Bellman equations

For type-A:

$$V_A(z) = \max\{\Pi_A(z;w) + \beta E[V_A(z') \mid z, A],\ -\kappa_{AB} + \Pi_B(z;p_m,\tau_{ext}) + \beta E[V_B(z') \mid z, B],\ -\kappa_{AC} + \Pi_C(z;p_m$$

For type-C:

$$V_C(z) = \max\{\Pi_C(z;p_m,w) + \beta E[V_C(z') \mid z, C],\ -\kappa_{CA} + \Pi_A(z;w) + \beta E[V_A(z') \mid z, A],\ -\kappa_{CB} + \Pi_B(z;p_m,\tau$$

Thus τ_{ext} also affects A and C values *indirectly*, through how it changes the attractiveness of the B organizational state.

6.3 Entry condition

Let the value of entering as type-B be:

$$V_B^{entry} = \int V_B(z_0)\, dG(z_0) - c_{EB}.$$

Then lower τ_{ext} raises type-B entry value:

$$\tau_{ext} \downarrow \Rightarrow V_B(z) \uparrow \Rightarrow V_B^{entry} \uparrow .$$

So the governance-cost mechanism affects both incumbent behavior and the entry margin.

7 What is identified in the data

7.1 Directly observed objects

The model primitive τ_{ext} is not directly observed. What can be directly observed are product-level outcomes that are shaped by it.

The most important variables are:

1. **External commercialization adoption**

   $$\chi_{ipt}^{ext} = 1\{\text{holder}_{ipt} \neq \text{manufacturer}_{ipt}\}.$$

   This captures whether the product is commercialized through separated holder–manufacturer roles.

2. **Type-B external commercialization**

   $$\chi_{ipt}^{B\to ext} = 1\{\text{B-type holder uses an external qualified manufacturer}\}.$$

3. **B-originated conversion rate**

$$Conv_t^B = \frac{\#\{\text{B-originated original-drug products realized}\}}{\#\{\text{B-originated projects/applications}\}}.$$

4. **New original-drug products commercialized**

$$N_t = \#\{\text{new original-drug products commercialized at time } t\}.$$

7.2 Supply-side shifter for governance friction

Construct a pre-policy measure of qualified manufacturing capacity:

$$Cap_{r,d,0} = \#\{\text{qualified manufacturers in region } r \text{ and dosage/process class } d \text{ before MAH}\}.$$

This variable is important because if MAH reduces the governance cost of external commercialization, its effect should be larger where a suitable external manufacturing partner is easier to find and organize.

8 Empirical implementation

8.1 Core reduced-form regressions

Regression 1: route adoption

$$\chi_{ipt}^{B\to ext} = \alpha_i + \gamma_t + \beta Post_t + \delta X_{ipt} + \varepsilon_{ipt}.$$

Expected sign:

$$\beta > 0.$$

Regression 2: conversion rate

$$Conv_{rdt}^B = \alpha_{rd} + \gamma_t + \beta Post_t + \varepsilon_{rdt}.$$

Expected sign:

$$\beta > 0.$$

Regression 3: heterogeneity by external capacity

$$Y_{rdt} = \alpha_{rd} + \gamma_t + \beta_1 Post_t + \beta_2 (Post_t \times Cap_{r,d,0}) + \varepsilon_{rdt},$$

where Y_{rdt} can be:

- $\chi^{B\to ext}$,
- $Conv^B$,

- N_t.

Expected sign:

$$\beta_2 > 0.$$

This is the sharpest reduced-form implication of the mechanism: if MAH lowers the governance friction of external commercialization, then the reform should have larger effects where the relevant external manufacturing route is easier to organize.

8.2 Data implementation plan

The preferred sequence is:

1. Build a product-year dataset containing:
   - product approval number,
   - drug name, dosage form, and therapeutic category,
   - original-drug indicator,
   - holder/MAH identity,
   - manufacturer identity,
   - approval/marketing timing,
   - province/city,
   - manufacturer qualification scope.
2. Construct direct route-adoption variables:

$$\chi_{ipt}^{ext}, \quad \chi_{ipt}^{B\to ext}.$$

3. Construct implementation proxies:

$$Conv_t^B, \quad N_t.$$

4. Add the pre-policy supply-side shifter:

$$Cap_{r,d,0}.$$

5. Use firm annual reports and prospectuses only as supplementary validation for keywords such as:
   - entrusted manufacturing,
   - contract manufacturing,
   - MAH/holder,
   - product release,
   - CMO/CDMO.

9 How τ_{ext} is calibrated

τ_{ext} should not be "directly estimated" from one variable. Instead, it should be disciplined by a set of moments:

1. external commercialization adoption rate;
2. type-B external commercialization share;
3. B-originated conversion rate;
4. number of new original-drug products;
5. heterogeneity with respect to $Cap_{r,d,0}$;
6. producer-side outcomes of qualified manufacturers.

Thus the model treats τ_{ext} as a latent structural friction that is pinned down by:

- route adoption,
- route productivity,
- route capacity heterogeneity.

10 Bottom line

The updated structure can be summarized in one chain:

$$MAH \Rightarrow \tau_{ext} \downarrow \Rightarrow H_B^{out} \uparrow \Rightarrow \Psi_B \uparrow \Rightarrow x_B \uparrow,\ V_B \uparrow,\ Entry_B \uparrow,\ N_t \uparrow .$$

This has two advantages.

First, it gives MAH a single, sharp, and institutionally grounded entry point in the model.

Second, it aligns the theory with the empirical objects that can actually be measured: route adoption, type-B conversion, new original-drug products, and heterogeneity by external qualified manufacturing capacity.